PES UNIVERSITY

Software Engineering

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5-F

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	**The system shall evaluate user targeting rules**, such as User ID hash and Beta cohort membership, and return the correct boolean feature flag state.	High	**Pass:** A user belonging to a configured 10% rollout cohort receives an enabled flag state, while a user outside the cohort receives a disabled flag state. **Fail:** A user receives a flag state inconsistent with the configured targeting rules.	Accurate feature flag evaluation is the core functionality of the system.
FR-002	Functional	**The system shall allow a Software Engineer to create, update, enable, disable, and delete feature flags.**	High	**Pass:** The Software Engineer can create a feature flag with a unique identifier and successfully update its state or configuration. **Fail:** The requested feature flag operation is not saved or the system allows duplicate identifiers.	Feature flags must be managed throughout their lifecycle.
FR-003	Functional	**The system shall allow a Release Manager to configure percentage-based rollouts for a feature flag.**	High	**Pass:** When a rollout percentage of 10% is configured, the system enables the feature only for users selected by the configured cohort evaluation mechanism. **Fail:** Users outside the selected rollout cohort incorrectly receive the enabled feature.	Percentage-based rollouts enable gradual and controlled feature releases.
FR-004	Functional	**The system shall allow users to define environment-specific**	Medium	**Pass:** When different configuration values are defined for development and production, a client	Environment overrides allow different configurations to

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		configuration overrides for development, staging, and production environments.		requesting a value for each environment receives the correct environment-specific value. **Fail:** A client receives a configuration value belonging to another environment.	be safely used across deployment environments.
FR-005	Functional	**The system shall synchronize updated feature flag and configuration values with connected client SDKs.**	High	**Pass:** After a feature flag or configuration is updated, a connected client SDK receives the updated value without manual configuration changes. **Fail:** The SDK continues using an outdated value after synchronization.	Clients need current feature and configuration data to behave correctly.
NFR-001	Non-Functional — Performance	**The feature flag evaluation API shall execute in under 15 ms under simulated peak load.**	High	**Pass:** Performance tests show that feature flag evaluation completes in less than 15 ms under the defined simulated peak load. **Fail:** Evaluation takes 15 ms or longer.	Feature flag evaluation may occur on critical application paths, so overhead must remain minimal.
NFR-002	Non-Functional — Security	**The system shall allow only authorized users to create, modify, enable, disable, or delete feature flags and configuration overrides.**	High	**Pass:** Authorized users can perform permitted operations, while unauthorized users receive an access-denied response and no configuration change occurs. **Fail:** An unauthorized user successfully modifies a feature flag or configuration.	Unauthorized changes could affect application behavior, production stability, and software releases.